

Computer System Architecture — Master Notes

Compiled from whiteboard lectures, classroom discussions, and deep-dive sessions.
Topics: Computation Hardware → Memory Hierarchy → Networking → Cloud Architecture

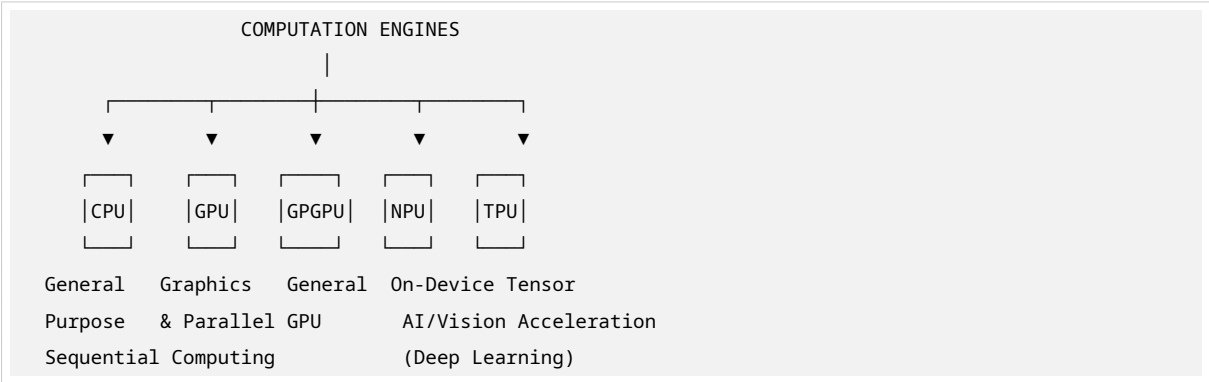
Table of Contents

- 1. Computation & Specialized Processing Hardware
- 2. The Abstraction Stack
- 3. Von Neumann Architecture & The CPU
- 4. Memory & Storage Hierarchy
- 5. Networking, IP Addressing & Packet Routing
- 6. Full-Stack Web Architecture & Cloud Virtualization
- 7. Integrated Story: How It All Connects
- 8. Keyword Glossary

1. Computation & Specialized Processing Hardware

1.1 The Five Processing Engines

Modern computing has moved beyond single general-purpose CPUs to specialized accelerators designed for parallel and AI/ML workloads.



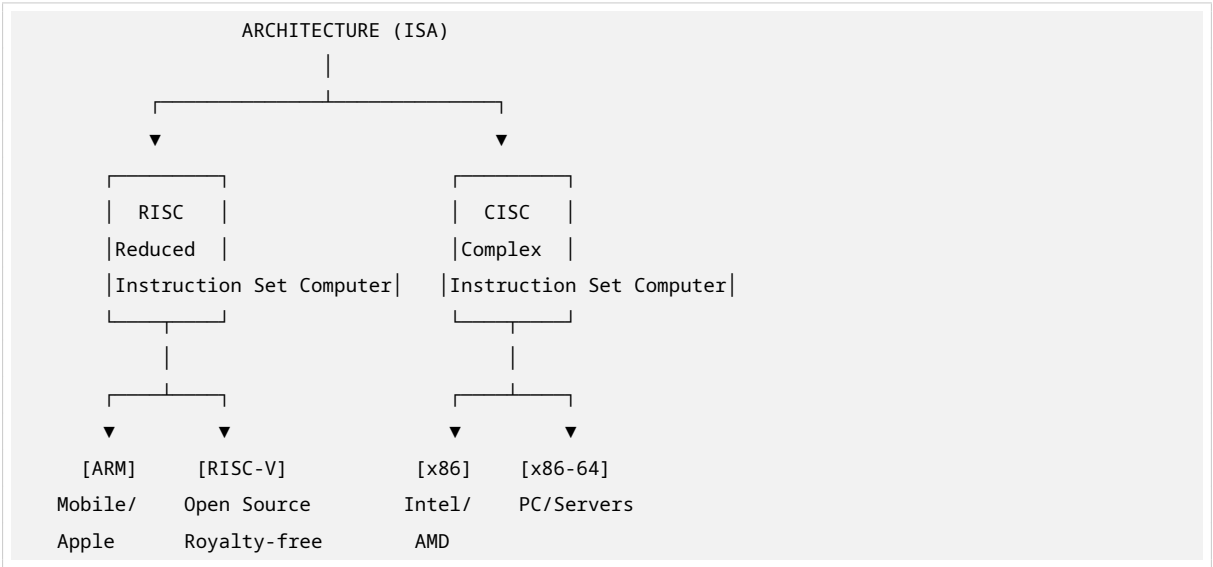
Processor	Cores	Best For	Weakness	Analogy
CPU	4-16 powerful	Complex, varied sequential tasks	Slow at parallel tasks	Brain surgeon
GPU	1,000-10,000 simple	Graphics, parallel math (SIMD)	Bad at branching logic	Factory assembly line
GPGPU	Same as GPU	Non-graphics parallel compute (AI training)	Same as GPU	Factory doing new jobs
NPU	Hundreds	On-device AI (face unlock, voice)	Only AI inference	Airport security guard
TPU	Thousands of matrix units	Massive AI training	Only matrix math	Industrial printing press

SIMD = Single Instruction, Multiple Data. One command operates on thousands of data points simultaneously.

Mat Mul (Matrix Multiplication) = The core operation of every neural network. CPUs do it sequentially; GPUs parallelize it; TPUs execute entire matrices in hardware using **systolic arrays**.

1.2 Processor Architectures: RISC vs CISC

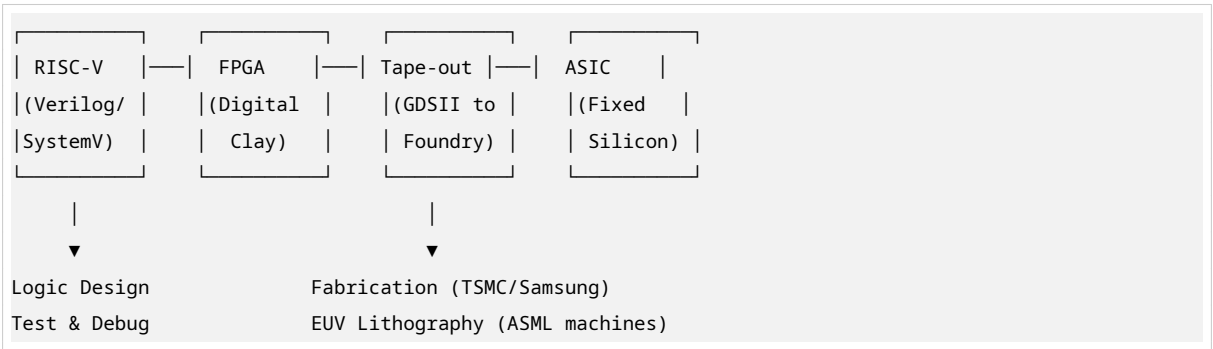
The **ISA (Instruction Set Architecture)** defines the vocabulary of commands a processor understands.



Feature	CISC	RISC
Philosophy	Complex hardware, simple software	Simple hardware, complex software
Instructions	Few, multi-step	Many, single-cycle
Power	Higher	Lower
Examples	Intel, AMD (x86/x86-64)	ARM (phones, Apple M), RISC-V

RISC-V is open-source and royalty-free. Anyone can design processors using it without licensing fees.

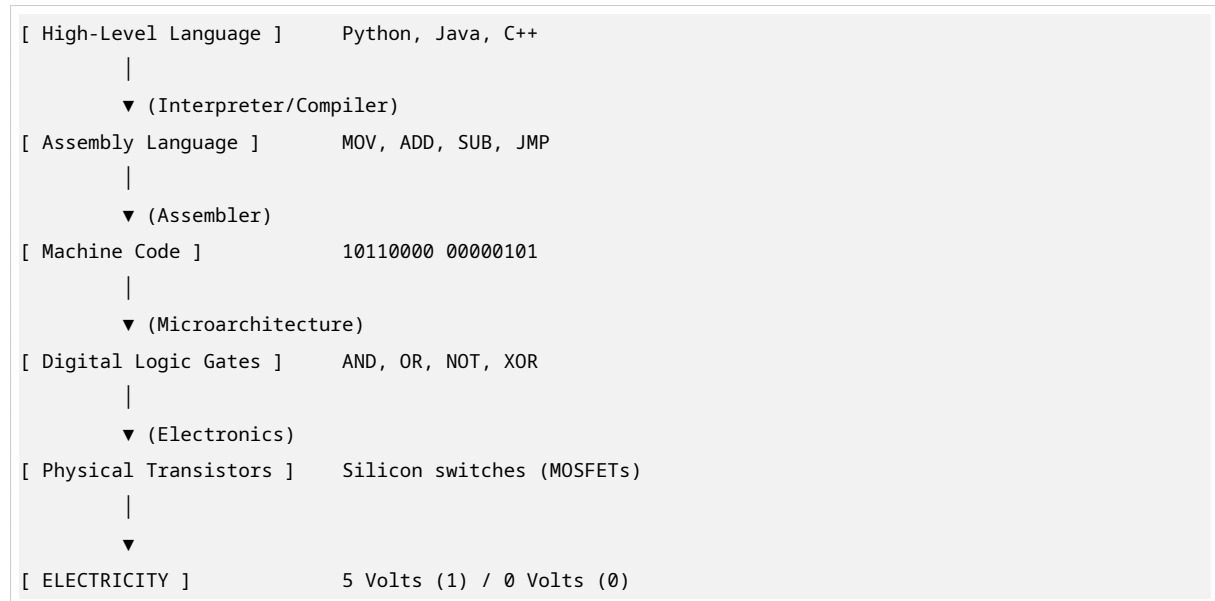
1.3 Custom Silicon Pipeline: From Design to ASIC



- **FPGA** = Field-Programmable Gate Array. Reconfigurable “digital clay” for testing before spending millions on manufacturing.
- **Tape-out** = Final design sent as GDSII file to a foundry.
- **ASIC** = Application-Specific Integrated Circuit. Fixed, non-reprogrammable, maximum efficiency.
- **EUV Lithography** = Extreme Ultraviolet light etching patterns as small as 3nm (about 30 atoms wide).

2. The Abstraction Stack

Every piece of code you write must eventually become electricity flowing through silicon.



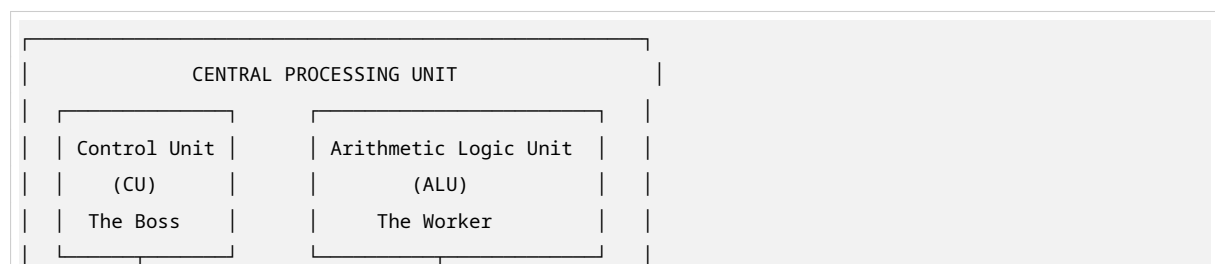
Abstraction means each layer hides the complexity below it. A Python programmer never thinks about transistors. A chip designer never thinks about Python.

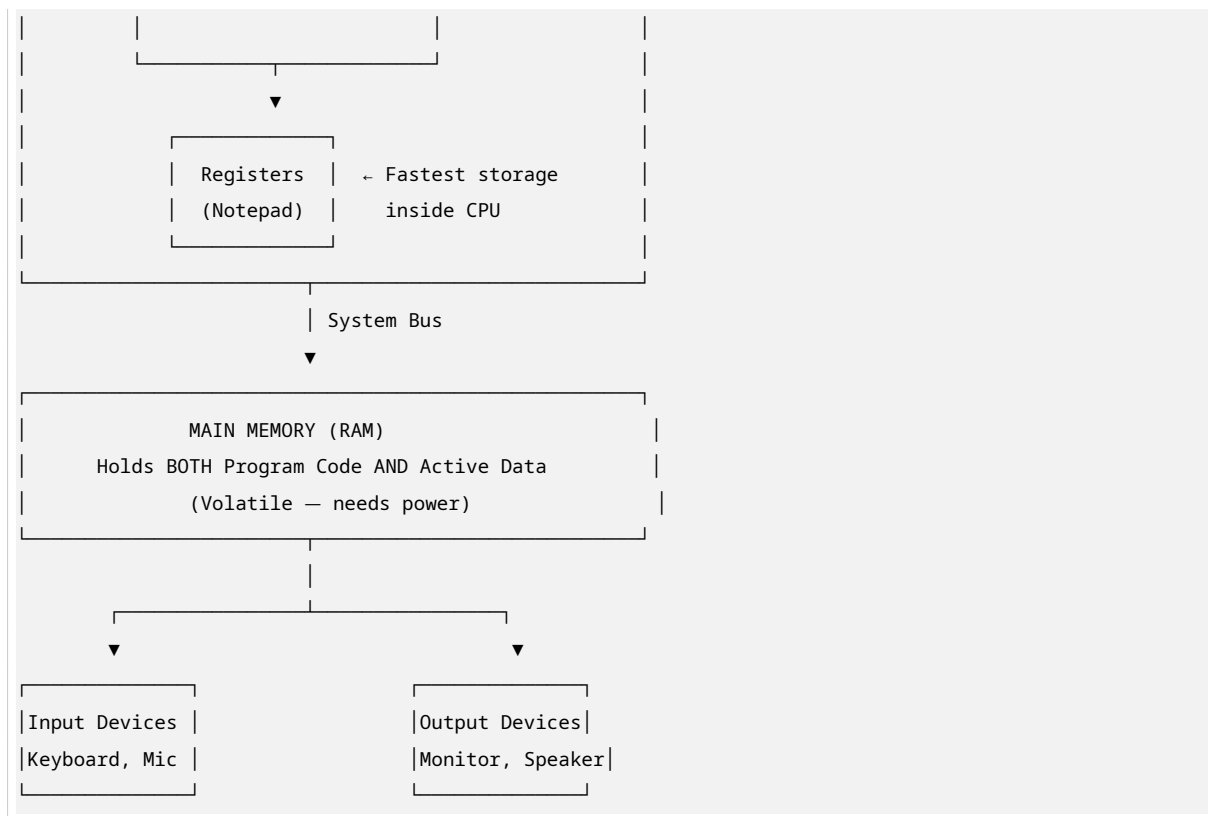
CMOS (Complementary Metal-Oxide-Semiconductor) = Modern transistor technology where paired transistors ensure almost **zero power consumption when stable** (not switching). Power is only consumed during state transitions.

3. Von Neumann Architecture & The CPU

3.1 The Stored-Program Concept

Proposed by John von Neumann (1945). Both **program instructions** and **data** share the same physical memory space. This is why your phone can run WhatsApp one minute and Instagram the next.





3.2 CPU Components

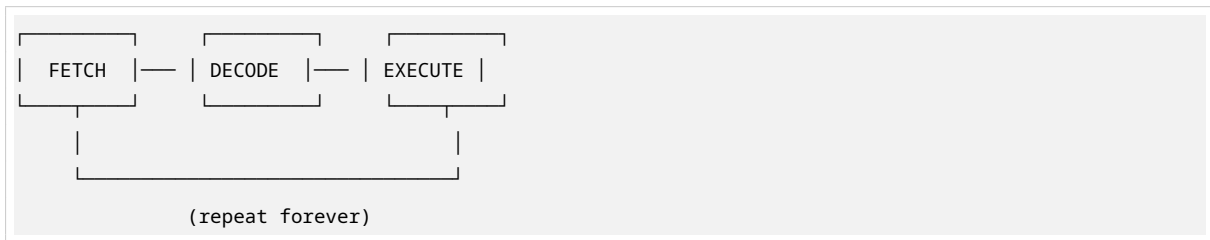
Component	Full Name	Role	Analogy
CU	Control Unit	Fetches, decodes, directs instructions	Traffic police / Mayor
ALU	Arithmetic Logic Unit	Performs math (+, −, ×, ÷) and logic (AND, OR, compare)	Accountant
Registers	—	Tiny, super-fast storage inside CPU	Writing on your hand

3.3 Key Registers

Register	Holds	Analogy
PC	Address of NEXT instruction	Finger pointing to next recipe step
IR	CURRENT instruction being decoded	Sentence being read aloud
ACC	Result of last calculation	Answer just written down
MAR	Memory address to access	“Go to shelf 5”
MDR	Data being transferred	The book from the shelf

3.4 The Fetch-Decode-Execute Cycle

The **only** thing a CPU ever does. Repeats billions of times per second.



Fetch: PC → MAR → Read signal → RAM puts instruction on Data Bus → MDR → IR → PC increments

Decode: CU interprets the instruction in IR

Execute: ALU calculates or data moves; result written back if needed

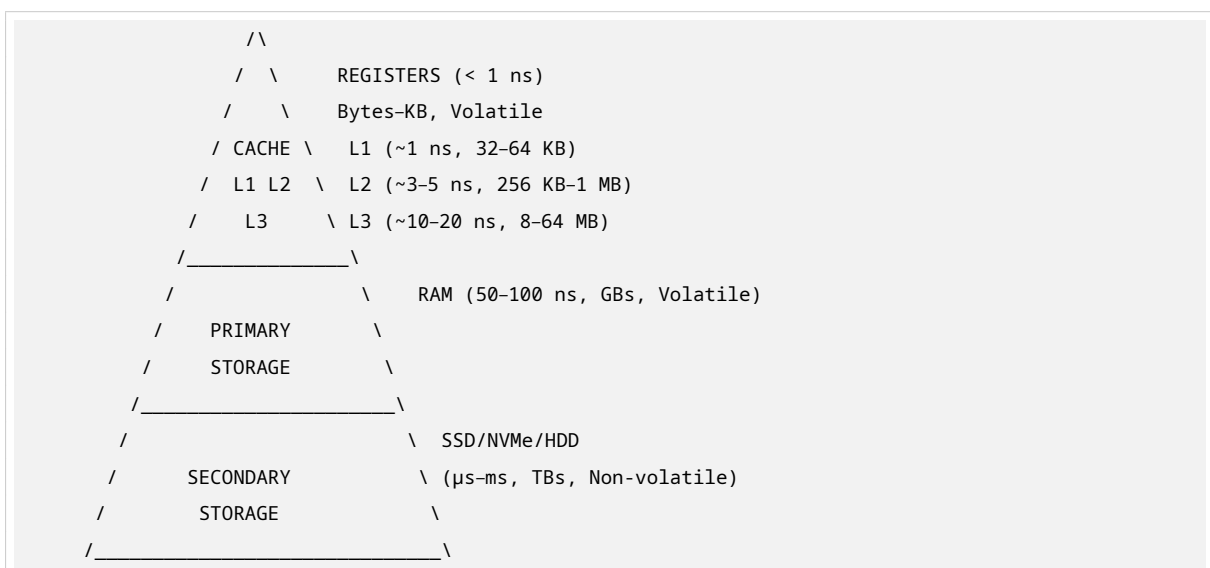
Clock Speed: 1 GHz = 1 billion cycles per second. A 3 GHz CPU does 3 billion fetches/decodes/executes per second.

3.5 The System Bus

Bus	Direction	Carries	Analogy
Address Bus	Unidirectional (CPU → RAM)	Memory address	“Go to shelf 5”
Data Bus	Bidirectional	Actual data/instructions	The book itself
Control Bus	Bidirectional	Read/Write signals, interrupts	“Read now!” / “Done!”

4. Memory & Storage Hierarchy

4.1 The Pyramid



4.2 Memory Types

Type	Speed	Size	Volatile?	Use
Registers	< 1 ns	Bytes–KB	Yes	Inside CPU, temporary holding
L1 Cache	~1 ns	32–64 KB	Yes	Per core, most frequently used
L2 Cache	~3–5 ns	256 KB–1 MB	Yes	Per core or shared
L3 Cache	~10–20 ns	8–64 MB	Yes	Shared across all cores
RAM	50–100 ns	GBs	Yes	Active programs and data
SSD	~100 μ s	100s GB–TB	No	Fast permanent storage
NVMe	~10–50 μ s	100s GB–TB	No	SSD via PCIe, very fast
HDD	~5–10 ms	TBs	No	Cheap, slow, moving parts

4.3 Key Concepts

Cache Hit: Data found in cache $\rightarrow$ fast access.

Cache Miss: Data not in cache $\rightarrow$ must go to RAM $\rightarrow$ slow.

Locality of Reference: Programs tend to access nearby data next. CPU fetches entire blocks into cache.

Volatile: Needs power (RAM, cache, registers).

Non-volatile: Retains data without power (SSD, HDD, Flash, ROM).

Why is RAM volatile? It stores bits as electrical charge in capacitors. Capacitors leak charge like a bucket with a hole and must be refreshed thousands of times per second. When power dies, charge drains in milliseconds.

Flash Memory (SSD): Uses transistors that trap electrons in a floating gate. Electrons stay trapped without power = non-volatile.

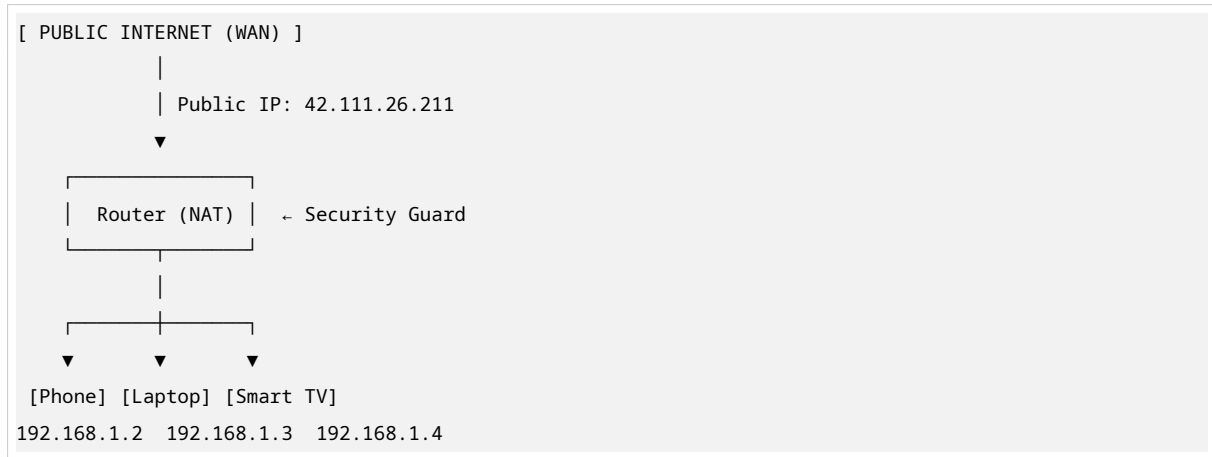
5. Networking, IP Addressing & Packet Routing

5.1 Public IP vs Private IP

Type	Definition	Scope	Assigned By	Analogy
Public IP	Globally unique address	Internet (WAN)	ISP	Building' s street address

Table 7 – continued

Type	Definition	Scope	Assigned By	Analogy
Private IP	Internal network address	Local network (LAN)	Router	Flat number inside building



NAT (Network Address Translation): The router replaces private IPs with the public IP on outgoing traffic and does the reverse for incoming replies.

5.2 IPv4 Math

- **32 bits** divided into 4 octets (8 bits each)
- Each octet ranges 0–255
- Total addresses: $2^{32} = 4,294,967,296$ (~4.3 billion)
- **Problem:** Exhausted. 8+ billion people, multiple devices each.

5.3 Private IP Ranges (IANA Reserved)

Range	CIDR	Common Use
10.0.0.0 – 10.255.255.255	/8	Large enterprises, cloud (AWS VPC)
172.16.0.0 – 172.31.255.255	/12	Medium networks, Docker
192.168.0.0 – 192.168.255.255	/16	Home routers, small offices

5.4 IPv6

- **128 bits**, written in hexadecimal with colons
- Total addresses: 2^{128} — enough for every grain of sand on Earth
- Adoption is slow because IPv4 + NAT still works; replacing global infrastructure takes decades

5.5 Network Security Devices

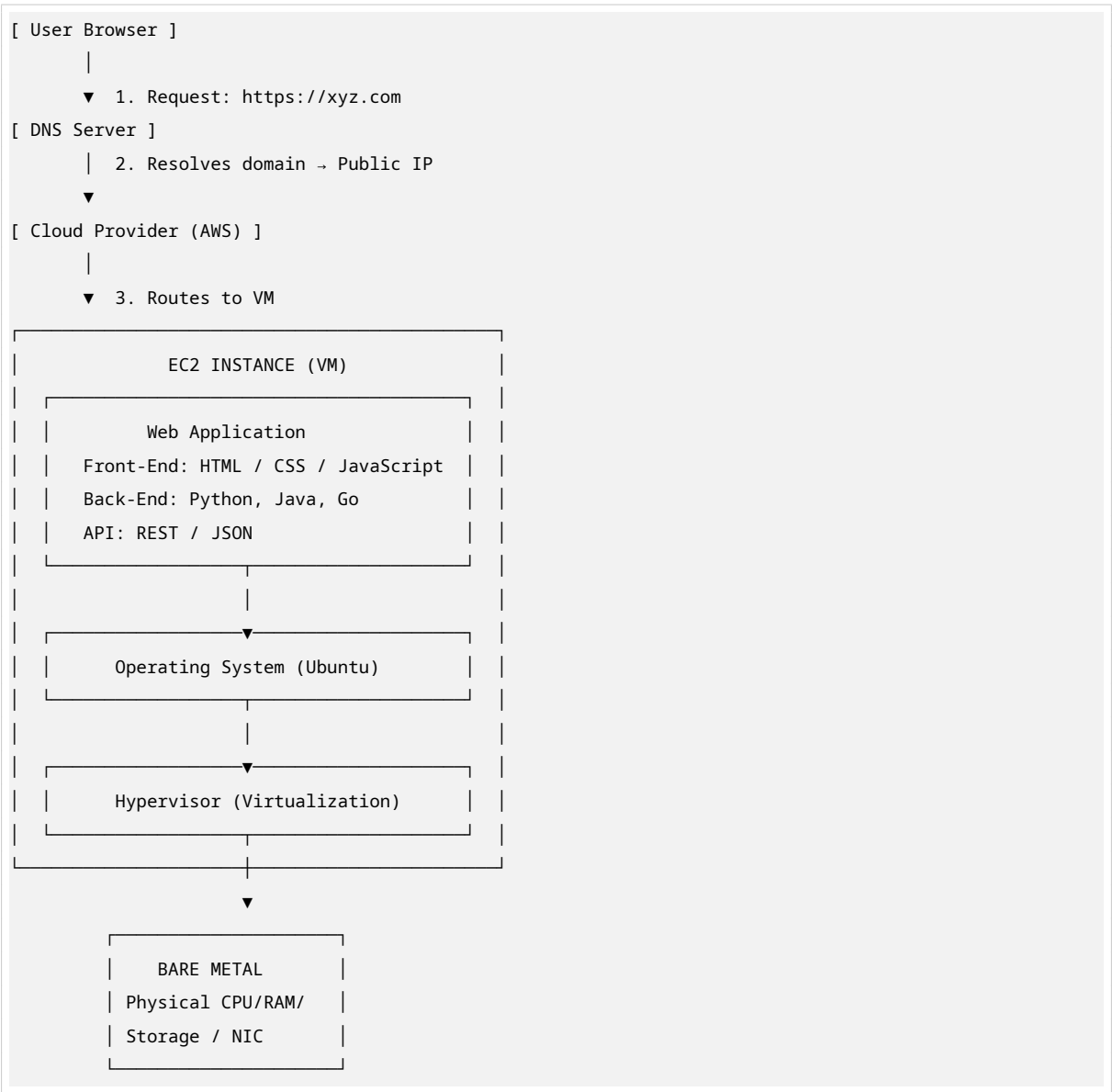
Device	Function	Layer	Analogy
Switch	Connects devices within LAN using MAC addresses	Layer 2	Internal mail sorter

Table 9 – continued

Device	Function	Layer	Analogy
Router	Connects LAN to WAN using IP addresses	Layer 3	Building gate guard
Firewall	Filters traffic by security rules	—	Bouncer with guest list
IDS	Detects threats, alerts only (passive)	—	CCTV camera
IPS	Detects threats, blocks actively	—	Guard with baton

6. Full-Stack Web Architecture & Cloud Virtualization

6.1 The Web Stack



6.2 Key Terms

Term	Definition	Analogy
Front-End	What user sees (HTML, CSS, JS)	Restaurant menu
Back-End	Server logic and databases	Kitchen behind the door
API / REST API	Communication protocol between layers	Waiter carrying orders
DNS	Domain Name System; name → IP	Phonebook
Bare Metal	Physical server, OS runs directly on hardware	Standalone house
VM / EC2	Software-emulated computer	Apartment in a building
Hypervisor	Manages multiple VMs on one physical server	Building manager
SSH	Secure Shell; encrypted remote login (Port 22)	Secure phone line into server
Data Center	Facility with thousands of servers	Giant apartment complex
Hyperscaler	AWS, Azure, GCP	Real estate giants of tech

6.3 AWS Free Tier Billing

- **750 hours/month** of t2.micro or t3.micro for 12 months = FREE
- Meter ticks **only while instance is RUNNING**
- **Stop Instance** = turn off taxi meter; data preserved, no compute charges
- **Terminate** = delete permanently

7. Integrated Story: How It All Connects

Scenario: You unlock your phone and send a WhatsApp message.

1. **NPU** verifies your face locally (low-power AI)
2. **Front-End** (app) captures your tap
3. **API** packages message as JSON
4. **Private IP** (192.168.1.x) sends packet to **Router**
5. **NAT** swaps Private IP → **Public IP**
6. Packet travels via internet **routers** and **switches**
7. Reaches **Data Center** → **Hypervisor** → **VM** on **Bare Metal**
8. Server **CPU** runs **Fetch-Decode-Execute** to store message
9. Code path: **Python** → **Assembly** → **Machine Code** → **Logic Gates** → **Transistors**
10. Message stored in **RAM** temporarily, then **SSD** permanently
11. Recipient receives it via reverse path

Every topic connects. The abstraction stack, the CPU cycle, memory hierarchy, networking, and cloud architecture are all part of one continuous system.

8. Keyword Glossary

Term	Definition
Abstraction	Hiding lower-layer complexity; using upper layers without knowing what's below

Table 11 – continued

Term	Definition
Accumulator (ACC)	Register holding the result of ALU operations
Address Bus	Unidirectional bus carrying memory addresses from CPU
ALU	Arithmetic Logic Unit; performs math and logic operations
ASIC	Application-Specific Integrated Circuit; fixed final chip
Bare Metal	Physical server hardware with OS running directly
Binary	Base-2 number system using only 0 and 1
Bus	Set of wires connecting computer components
Cache	Fast buffer memory between CPU and RAM
CISC	Complex Instruction Set Computer (Intel, AMD x86)
CMOS	Complementary MOS; near-zero power when stable
Control Bus	Carries command and timing signals
Control Unit (CU)	Fetches and decodes instructions; the manager
CPU	Central Processing Unit; the brain
Data Bus	Bidirectional bus carrying actual data
De Morgan's	Theorems for converting between AND/OR with complements
DRAM	Dynamic RAM; capacitor-based, needs refresh
EUV Lithography	Extreme ultraviolet light etching for 3nm chips
Fetch-Decode-Execute	The only cycle a CPU runs
Firewall	Filters network traffic by rules
FPGA	Field-Programmable Gate Array; reconfigurable test chip
GPU	Graphics Processing Unit; massive parallel cores
HDD	Hard Disk Drive; spinning magnetic platters
Hypervisor	Software managing multiple VMs
IDS/IPS	Intrusion Detection/Prevention System
IP Address	Network device identifier
IPv4/IPv6	Internet Protocol versions 4 and 6
ISA	Instruction Set Architecture
LAN	Local Area Network
MAR	Memory Address Register
MDR	Memory Data Register
Memory Hierarchy	Speed-size-cost pyramid of storage
NAT	Network Address Translation
Non-volatile	Retains data without power
NPU	Neural Processing Unit
PC	Program Counter
PMIC	Power Management IC; regulates voltage for phone chips
Private IP	Internal LAN address
Public IP	Global internet address
RAM	Random Access Memory; volatile primary storage
Register	Fast storage location inside CPU
RISC	Reduced Instruction Set Computer
RISC-V	Open-source ISA
Router	Connects networks at Layer 3
SIMD	Single Instruction Multiple Data

Table 11 – continued

Term	Definition
SRAM	Static RAM; flip-flop based, no refresh
SSD	Solid State Drive; flash memory
Switch	Connects devices within LAN at Layer 2
System Bus	Address + Data + Control buses
Tape-out	Sending final chip design to factory
TPU	Tensor Processing Unit
Transistor	Electronic switch; fundamental building block
Von Neumann	Stored-program computer architecture
Volatile	Loses data without power
WAN	Wide Area Network (the internet)

“A computer is just a box of switches turning on and off at the speed of light. Everything else is abstraction.”